17. Simple Random Search

February 23, 2026 8:53 AM

What do we do in Statistics, Data Sci, ML, etc?

→ Fit Models to Data?

→ Optimizing Something.

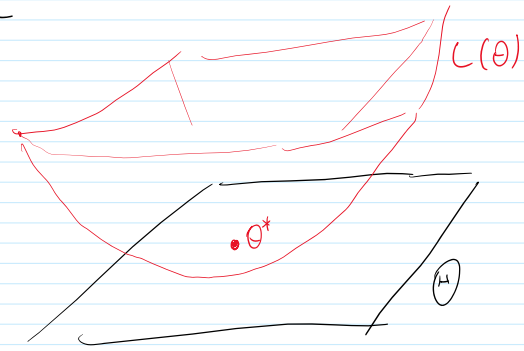
- Maximize a likelihood! (Freq)
- Max a posteriori (Bayesian)
- Minimize → Loss
  → Risk
  → Regret

Goal: $\theta^* = \arg\min_\theta \{L(\theta)\}$ ← Find the vector θ^* that minimizes some Loss function.

$\theta \in \mathbb{R}^d$, $L: \mathbb{R}^d \to \mathbb{R}$

Noise-Free or Noisy Setting

Do we obs $L(\theta)$ or $L(\theta) + \varepsilon$ ← added Noise



Simple Random Search

Require:
- Loss Function $L(\theta)$
- Domain Θ
- # of iterations

also need Θ to be Bounded

Algorithm:
(1) pick a uniformly random $\theta \in \Theta$
(2) Compute $L(\theta)$
(3) repeat this n times and keep the "best" θ, i.e. θ that gives the smallest Loss value.

* this will eventually find θ^* as $n \to \infty$ but doesn't use any info about $L(\theta)$ except evaluating it n times.

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→ the sequence of θ's will converge (in some sense) to θ^* under some conditions

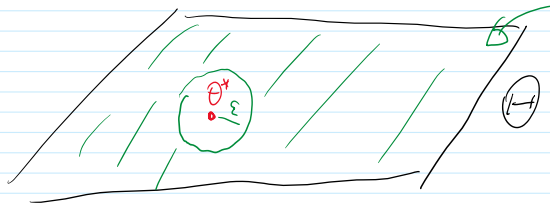
→ θ^* is unique and $\forall\, \varepsilon > 0$, $\inf_{\|\theta - \theta^*\| > \varepsilon} L(\theta) > L(\theta^*)$

→ $L(\theta^*) > -\infty$

→ "Continuity": $\forall\, \varepsilon > 0\ \exists\, \delta > 0$ s.t.

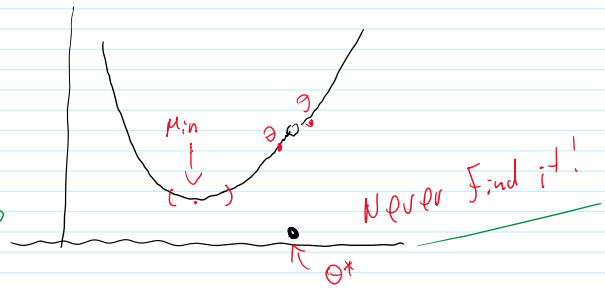
What's the point of this?? ← $P(L(\theta) < L(\theta^*) + \varepsilon) \geq \delta$

outside of some local region, $L(\theta)$ is bigger than $L(\theta^*)$

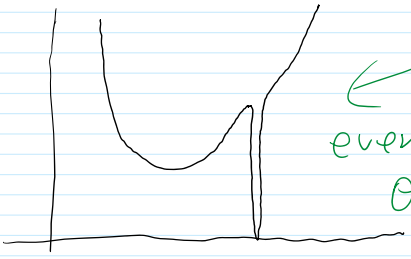


Outside of this ε-ball, $L(\theta)$ is bounded away for the minimum.

For "continuity", some positive probability of getting close to the optimum



Other poorly constructed losses



eventually find θ^* but n would be very very big

Going forward, we will consider actual interesting optimization algorithms, but there's a trade off between knowledge about $L(\cdot)$ and simplicity of the algorithms

→ evaluate $L(\cdot)$

→ know the gradient $\nabla L(\cdot)$

→ can we estimate $\nabla L(\cdot)$

→ How many evaluations do we need?